Separable equations

We saw that first order linear equations are differential equations in the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

In contrast, **first order separable differential equations** are equations in the form

$$N(y)\frac{dy}{dx} = M(x)$$

$$N(y)y' = M(x)$$

We call these “separable” equations because we can separate the variables onto opposite sides of the equation. In other words, we can put the x terms on the right and the y terms on the left, or vice versa, with no mixing.

Notice that the two previous equations above representing a separable differential equation are identical, except for the derivative term, which is represented by dy/dx in the first equation and by y' in the second equation.

When we’re working with separable differential equations, we usually prefer the dy/dx notation (Leibniz notation), because this notation makes it easier to see how to separate variables. Starting with dy/dx, we move the dx notation to the right, leaving just the dy notation on the left:

$$N(y)\ dy = M(x)\ dx$$

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How to solve separable equations

Most often, we'll be given equations in the form

$$N(y)\frac{dy}{dx} = M(x)$$

Here's the standard process we'll follow to find a solution to this separable differential equation. First, we'll separate variables by moving the dx to the right side.

$$N(y)\ dy = M(x)\ dx$$

With variables separated, we'll integrate both sides of the equation.

$$\int N(y)\ dy = \int M(x)\ dx$$

Normally when we evaluate an indefinite integral, we need to add in a constant of integration. So after integrating, we would expect to have something like

$$\mathbf{N}(y) + C_1 = \mathbf{M}(x) + C_2$$

where $\mathbf{N}(y)$ is the integral of $N(y)$, and $\mathbf{M}(x)$ is the integral of $M(x)$. But when we solve for $\mathbf{N}(y)$, we get

$$\mathbf{N}(y) = \mathbf{M}(x) + C_2 - C_1$$

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What we have to realize here is that, if C_2 and C_1 are constants, then $C_2 - C_1$ is also a constant. So we can make a substitution and represent $C_2 - C_1$ as just one simple constant C.

$$\mathbf{N}(y) = \mathbf{M}(x) + C$$

For this reason, we don't need to bother adding constants to both sides of the equation when we integrate. Instead of integrating to $\mathbf{N}(y) + C_1 = \mathbf{M}(x) + C_2$, we can always skip these intermediate steps and go straight to $\mathbf{N}(y) = \mathbf{M}(x) + C$.

Once we have the equation in this form, the goal will be to express y explicitly as a function of x, meaning that our solution equation is solved for y, like $y = f(x)$. Sometimes we won't be able to get y by itself on one side of the equation, and that's okay. If we can't, we'll just settle for an implicit function, which means that y and x aren't separated onto opposite sides of the solution.

In summary, our solution steps will be:

1. If necessary, rewrite the equation in Leibniz notation.
2. Separate variables with y terms on the left and x terms on the right
3. Integrate both sides of the equation, adding C to the right side
4. If possible, solve the solution equation specifically for y.

Let's work through an example so that we can see these steps in action.

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Example

Find the solution of the separable differential equation.

$$y' = y^2 \sin x$$

Let's write the equation in Leibniz notation, changing y' to dy/dx.

$$\frac{dy}{dx} = y^2 \sin x$$

Separate the variables, collecting y terms on the left and x terms on the right.

$$dy = y^2 \sin x \; dx$$

$$\frac{1}{y^2} \; dy = \sin x \; dx$$

With variables separated, and integrating both sides, we get

$$\int \frac{1}{y^2} \; dy = \int \sin x \; dx$$

$$\int y^{-2} \; dy = \int \sin x \; dx$$

$$-y^{-1} = -\cos x + C$$

$$-\frac{1}{y} = -\cos x + C$$

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$$\frac{1}{y} = \cos x + C$$

Notice how we just multiplied through the equation by -1, but we didn't change the sign on C. That's because keeping C is a little simpler than $-C$, and we'll still end up with the same solution equation either way.

Finally, solving for y gives the solution to the separable differential equation.

$$1 = y(\cos x + C)$$

$$y = \frac{1}{\cos x + C}$$

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